

Simple Machines — Lesson 2

ICSE Class 6 Physics · 10 MCQs · 1 mark each · Name: _____ Date: _____

Syllabus: Pulley and its mechanical advantage, wheel and axle, inclined plane, wedge, screw, care of machines.**Instructions:** Each question has one correct option. Tick or circle the letter of the best answer. Attempt all questions.**1. A pulley is a wheel with a grooved rim over which passes a:**

- A. chain only
- B. wooden plank
- C. metal rod
- D. rope or string

2. The ideal mechanical advantage of a single fixed pulley is:

- A. 2
- B. 1
- C. 0.5
- D. 4

3. A single fixed pulley is useful because it:

- A. changes the direction of the effort
- B. removes the load's weight
- C. creates energy
- D. doubles the effort

4. The ideal mechanical advantage of a single movable pulley is:

- A. 0.5
- B. 3
- C. 1
- D. 2

5. Which of these works as a wheel and axle?

- A. a steering wheel
- B. a knife
- C. a ramp
- D. a see-saw

6. In a wheel and axle, the mechanical advantage equals:

- A. load x effort
- B. length / height
- C. radius of wheel / radius of axle
- D. radius of axle / radius of wheel

7. A plank 5 m long is used to raise a load to a height of 1 m. The ideal MA of this inclined plane is:

- A. 5
- B. 0.2
- C. 4
- D. 6

8. Which of the following is an example of a wedge?

- A. a steering wheel
- B. a flagpole pulley
- C. an axe
- D. a see-saw

9. The spiral thread of a screw is an example of:

- A. a wheel and axle
- B. an inclined plane wrapped around a cylinder
- C. a pulley
- D. a lever

10. To keep a machine working efficiently we should:

- A. let it rust
- B. oil its moving parts regularly
- C. never clean it
- D. increase the friction

Answer Key

1. D

2. B

3. A

4. D

5. A

6. C

7. A

8. C

9. B

10. B